

CUSTOMER RETENTION DECISION REPORT

Should This Business Worry About Losing Customers, and What Should It Actually Do About It?

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Executive Summary

This report analyses churn across 1,000,000 customer records to answer a single question for the business: should you spend money trying to keep customers who look likely to leave and, if so, on whom? The short answer is that a flat, segment-wide discount does not pay for itself, no matter how it is targeted. A properly targeted, tightly budgeted offer loses far less money than a blanket one. The real opportunity is not a bigger discount but a cheaper, better-aimed intervention, tested properly before it is rolled out.

HEADLINE FINDING

Offering every High Risk customer a 15% retention discount would cost approximately £27 million over 12 months and only reduce churn from 10.7% to 9.8%, a net loss of roughly £54 million against doing nothing. Targeting the same offer at the 12,500 customers it is actually worth spending on, within a £2 million budget, cuts that loss to about £2.5 million: an 18× improvement in cost efficiency for a fraction of the spend.

The customers most worth saving are not simply the highest-risk ones. A model trained to predict who actually responds to being offered a discount, as opposed to who is merely likely to churn, finds that responsiveness is concentrated among newer, more dissatisfied, complaint-heavy customers and is close to zero among long-tenured, satisfied ones who were never going to leave regardless. Spending on the latter group is money wasted twice over.

This is consistent with independent industry research, not just this dataset: published studies on telecom retention warn that reflexive discounting trains customers to threaten cancellation for a reward and that discount effectiveness declines as prices rise. Personalised, well-targeted interventions are reported to outperform generic discounting by a meaningful margin. Our recommendation follows the same logic: stop offering blanket discounts and pilot a smaller, better-targeted intervention with a properly sized test before committing further budget.

1. The Business Question

Customer churn at this business currently sits at 9.9% overall, but that figure hides wide variation. Month-to-month customers churn 4.7× more often than two-year contract holders. One identifiable segment of about 192,000 customers (“High Risk”) churns at 14%, nearly half again the company average. The question this report answers is not “who is at risk”, which is well understood, but “what, if anything, should be done about it and does it actually pay for itself.”

2. What the Data Shows

2.1 Contract type is the dominant churn driver

Customers on rolling month-to-month contracts churn at 26.5%, against 12.7% for one-year and 5.7% for two-year contracts. This single factor separates customers more sharply than any other variable in the dataset.

Figure 1. Churn rate by contract type.

2.2 Four behavioural segments, statistically validated

Unsupervised clustering identifies four customer segments (Loyal Core, Stable, Watch List and High Risk) that separate primarily on complaints and service-call volume, not on how long someone has been a customer. A statistical test (silhouette score) suggests

two broader groups would separate more cleanly than four; we chose four regardless, because it gives the business more operational detail to act on, at a modest, disclosed cost in statistical tidiness.

A logistic regression confirms which factors drive churn and by how much: each additional complaint raises a customer's monthly cancellation odds by roughly 20%, as does each late payment, while each point of satisfaction (on a 1–9 scale) lowers it by around 10%. Tested on 250,000 customers the model had never seen, it correctly ranks a churner as riskier than a non-churner 67% of the time and, more usefully, its predicted probabilities are well calibrated: when it says a customer has a 15% chance of leaving, close to 15% of similar customers actually do.

Figure 2. Standardised effect of each factor on churn odds.

Figure 3. Predicted vs. observed churn rate, held-out test data. Points close to the diagonal indicate the model's probabilities can be trusted at face value.

3. Who Actually Responds to an Offer

Predicting who is at risk of churning and predicting who would actually be retained by a discount are different questions. Conflating them is the most common reason retention campaigns lose money. We built a treatment-effect (“uplift”) model to separate the two: it estimates, per customer, how much a discount would reduce their individual churn probability, rather than simply their baseline risk.

Because this dataset contains no historical discount experiment, the model was trained on a calibrated simulation: customers were randomly assigned a hypothetical discount, with the simulated effect size grounded in published research (10–15% relative churn reduction from targeted retention offers). We then tested whether the model could recover that planted effect from the data alone: the honest way to check whether the method works before trusting it on real decisions. It performed well. Predicted uplift correlates 0.94 with the true simulated effect. Customers the model ranks in the top 10% for predicted responsiveness show, on average, three times the true retention effect of those in the bottom 10%.

Figure 4. Model validation: customers predicted to respond most (right) show the largest true effect.

The customers the model identifies as most responsive look different from the customers who simply churn most. High-responsiveness customers average 10 months' tenure, 1.9 complaints and a 3.7/9 satisfaction score; low-responsiveness customers average 43 months' tenure, 0.4 complaints and an 8.1/9 satisfaction score. In plain terms: the people worth spending a retention budget on are recent, frustrated customers who are still deciding whether the business is worth staying with, not long-tenured, happy customers who were never at meaningful risk.

4. What Happens If You Act on This

We modelled three courses of action for the High Risk segment (192,064 customers) over a 12-month horizon: doing nothing, offering every customer a 15% discount and offering the same discount only to the customers the uplift model identifies as worth targeting, within a fixed £2,000,000 budget.

Scenario	Customers offered discount	Projected 12-month churn	Customers retained vs. baseline	Discount cost	Net impact vs. baseline
A: Do nothing	0	10.7%	0	£0	£0 (baseline)

Scenario	Customers offered discount	Projected 12-month churn	Customers retained vs. baseline	Discount cost	Net impact vs. baseline
B: Blanket discount	192,064	9.8%	+1,802	£27.4M	-£53.7M
C: Targeted (£2M budget)	12,487	10.6%	+267	£2.0M	-£2.5M

Figure 5. Net 12-month revenue impact of each scenario, relative to doing nothing.

Both intervention scenarios lose money against doing nothing, but the gap between them is the entire point. Blanket discounting spends £27.4 million to retain 1,802 extra customers, roughly £15,200 per customer saved, against an average annual customer value of under £1,000. Targeted discounting spends £2 million to retain 267 extra customers, still expensive, but eighteen times more cost-efficient, because the spend is concentrated on customers who are both likely to leave and likely to be swayed, rather than spread across everyone regardless of whether they needed convincing.

WHAT THIS MEANS FOR YOU

Under the assumptions tested here, a price discount does not pay for itself for this customer base at any scale we modelled. That is not a reason to do nothing. It is a reason to stop reaching for price as the retention lever and test a cheaper one instead. Section 6 sets out exactly how to do that properly.

5. If a Retention Budget Is Committed Anyway

If the business proceeds with a discount-based campaign regardless, the allocation should follow predicted cost-efficiency, not risk score alone. Ranking all eligible customers by expected return per pound of discount spent, then funding the best-ranked customers first until the budget is exhausted, recovers 8.4% of the money spent as retained revenue, compared with 2.8% for a blanket approach and 2.8% for a same-sized random selection. Targeting does not fix the underlying economics, but it is unambiguously the more disciplined way to spend the money if the spend is going to happen.

Figure 6. Cumulative net impact of spend, customers ranked by predicted cost-efficiency.

6. Recommendation and Suggested Next Step

Our recommendation is not to launch a discount-based retention campaign at the scale modelled here. Instead, we recommend piloting a lower-cost, non-price intervention (proactive outreach to newly dissatisfied, high-complaint customers, service recovery rather than a price concession) which published industry research associates with churn reductions in a similar range (10–15%) without the ongoing margin cost of a permanent discount.

Because that effect size is currently an assumption, not a measurement, it should be tested before being relied upon. A properly powered pilot, run as a randomised trial across a subset of the High Risk segment, would require approximately 14,900 customers per arm (around 29,700 in total) to detect a 9.2% relative reduction in churn at standard statistical confidence (80% power, 5% significance), or as few as 5,400 per arm if the true effect is closer to the upper end of published research (15%). Both are comfortably achievable within a segment of 192,000 customers and would give the business a measured, not assumed, answer before any wider rollout.

SUGGESTED NEXT STEP

Run a randomised pilot of the proposed intervention on $\approx 30,000$ High Risk customers (15,000 per arm) for one full billing cycle, measure the actual churn-rate difference and use that measured effect, not the assumption used in this report, to decide whether to scale it.

7. Confidence and Limitations

Every model in this report was validated on data it had not seen during fitting. Every headline figure is reported with its uncertainty rather than as a bare point estimate.

Churn model discrimination (AUC)	Churn model calibration	Survival model concordance
0.67	Strong	0.61

Figure 7. Bootstrap distribution (500 resamples) of the blanket-discount cost estimate; the range is narrow because the underlying customer base is large.

Three limitations are worth stating plainly. First, the uplift model's effect sizes are calibrated against published research, not measured from this business's own historical experiments, hence the recommendation to pilot before scaling. Second, the churn classifier is a moderate discriminator (AUC 0.67); it is far better used to rank relative risk and estimate probabilities than to make yes/no calls on individual customers. Third, the four-segment structure sits below the statistically “cleanest” split ($k=2$); it was chosen for business usefulness. That trade-off is disclosed here rather than hidden.

Appendix: Methodology and Sources

A.1 Data and methods

- 1,000,000-row customer churn dataset; cleaning, exploratory analysis and k-means segmentation (k=4, silhouette-validated) in Python (pandas, scikit-learn).
- Churn classifier: logistic regression, held-out test set (25%), evaluated by AUC-ROC, precision/recall and calibration.
- Survival modelling: Kaplan-Meier and Cox proportional hazards (lifelines), concordance index on held-out data.
- Uplift modelling: T-learner (two separately fitted logistic regressions, treated vs. control) on a calibrated simulated experiment; validated by correlation with the known simulated effect and decile-monotonicity.
- Budget allocation: greedy allocation by cost-efficiency ratio, the exact solution to the continuous relaxation of the underlying budget-constrained knapsack problem.
- Uncertainty: 500-sample non-parametric bootstrap on customer-level outcomes for headline financial estimates.
- Experiment design: two-proportion power calculation ($\alpha = 0.05$, power = 0.80) for the recommended pilot.

A.2 External sources referenced

- Bain & Company / Harvard Business Review: a 5-percentage-point increase in customer retention is associated with a 25–95% increase in profit.
- Multiple industry sources (Optimove, Invesp, HBR): acquiring a new customer typically costs 5–25× more than retaining an existing one.
- Industry research (2024–2025): personalised, targeted retention offers associated with approximately 15% relative churn reduction, versus approximately 10% for generic loyalty discounts.
- Telecommunications Policy (2025) and industry commentary: discount-based retention shows diminishing effectiveness as price rises and repeated reactive discounting can train customers to threaten cancellation for reward.

This report and the accompanying interactive dashboard were produced as a demonstration of end-to-end analysis: from raw data to a costed, validated business recommendation. For a project built on your own data, contact tamearonmullings@gmail.com.